Aditya Dalvi *Software Engineer*

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Pune, India

PROFILE

Backend Engineer with 3 years of experience specializing in Python, Node.js and JavaScript performance optimization, and AI tool integrations. Analytical problem-solver with a proven track record of resolving complex bottlenecks and deploying services on AWS (EC2, S3). Adept at leveraging modern development tools, including Docker, Git, and AI-assisted workflows, to deliver reliable cloud applications.

SKILLS

Languages
Python, SQL, JavaScript, HTML, CSS

Authentication & Security
JWT-based authentication, Role-Based Access Control (RBAC)

Frameworks
Django, Node.js, Fast API, Express

Architecture
REST APIs, MVC, Controller-Validator–Service–Repository pattern

Databases
PostgreSQL, MongoDB, SQLite

Cloud & DevOps
Git, Docker, GitLab, CI/CD pipelines, AWS (EC2, S3)

PROFESSIONAL EXPERIENCE

Backend Developer, Myelin 12/2024 – present | Pune, India

- Designed and developed scalable **REST APIs** using **Node.js, JavaScript**, **MongoDB**, and **Python** for multiple EdTech platforms, leveraging **Claude Code** to accelerate feature delivery, navigate complex codebase dependencies, and optimize schema design.
- Built a **RAG-based semantic retrieval system** using **LangChain** and **Weaviate** for an EdTech platform (Mytri), indexing curriculum content (questions, activities, examples) with multilingual embeddings and metadata filters (outcome, Bloom's level, difficulty, language).
- Designed and deployed AI-powered microservices using **FastAPI** and **OpenAI APIs**, implementing structured output parsing and robust prompt engineering to automate content generation reduced manual planning effort by **70%**
- Optimized AI API consumption and reduced latency by **65%** by implementing advanced caching layers, minimizing redundant **LLM calls** and managing rate limits
- Designed and implemented an asynchronous report card generation pipeline using **BullMQ** and **Redis**, queuing report-generation jobs per student.
- **Worked on Projects: HPC** (Holistic Progress Card), **Mytri** (A personal companion app), **Myelin** (A school management platform)

Associate Software Engineer, Augmented Transformations 01/2024 – 06/2024 | Pune, India

- Built a real-time analytics and reporting system for Computer Vision based CCTV safety monitoring using **Python**, **Django**, **React** and **Chart.js**, used by **100+** clients and reducing safety violations by **33%.**
- Optimized REST API performance by refining database queries and business logic, achieving up to a **90%** reduction in response times.
- Executed **data cleaning** and **transformation** workflows to remove inconsistencies and handle missing values, significantly improving data accuracy for analytics.

- Contributed to end-to-end delivery of computer vision solutions by testing object detection models and supporting production deployment for client use cases.

Trainee, Myelin 02/2023 – 10/2023 | Pune, India

- Built REST APIs for an Online Exam Portal capable of handling **5,000+ evaluations/month**, improving efficiency by **75%** and reducing errors by **40%**.
- Developed backend systems tracking **2,000+ students**, improving academic performance monitoring by **30%**.

PROJECTS

Network , Github 08/12/2022 – 27/12/2022

- **Built a Twitter-like social network web application.**
- Users can Create**,** Like, Edit, Delete posts. Reply to posts and View replies of a post. Users can follow and unfollow other users.
- Deployed this app on **AWS EC2**, utilizing **Amazon S3** for efficient storage and management of media and static files.
- **Developed using Python, Django, JavaScript, SQLite, HTML, CSS and Bootstrap.**

FlowGen 03/2026

- **Engineered** a high-performance web-based ER diagramming tool automating the visualization of complex PostgreSQL schema definitions into interactive relational maps.
- **Architected** an intelligent data generation engine capable of producing referentially consistent mock data for many-to-many relationships and recursive self-referencing hierarchies.
- **Developed** a custom SQL parser that identifies and visually badges database constraints including **Primary Keys (PK)**, **Foreign Keys (FK)**, **Unique (UQ)**, and **Check (CK)** constraints.
- **Developed using React and Pixi.js**

CERTIFICATES & COURSES

- CS50 Web Programming With Python and JavaScript - Harvard Extension
- Introduction to Programming with Python - Helsinki University
- CS50 Introduction to Computer Science - Harvard Extension

EDUCATION

Bachelor of Engineering, Savitribai Phule Pune University 2021 | Pune, India